

Andres Perez

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EDUCATION

University of Notre Dame

Notre Dame, IN

Bachelor of Science in Computer Engineering | Minor: Engineering Corporate Practice | GPA: 3.7

May 2028

Activities: Irish Sat GOAT Lab, Robotics Football Club Electrical Team, Incoming Career Assistant (Fall 2025)

Engineering Study Abroad: London, England

May. 2025 – Jun. 2025

WORK EXPERIENCE

University of Notre Dame

Notre Dame, IN

Student Researcher

Aug. 2024 – Present

- Researched under Prof. Chaoli Wang by data labeling image data and downloading GSV data of households, surveying heat dissipation in household.

ENGINEERING PROJECTS

Hardware-Accelerated Human Activity Recognition (HAR)

Long Beach, CA

Developer

- Built a wireless data acquisition system using two Arduino Uno R3s, an MPU-9250 IMU sensor, and nRF24L01 RF transceivers.
- Developed a C++ sketch to sample raw accelerometer and gyroscope data, format it, and transmit it wirelessly to a receiving station. Along with a Python script to receive, parse, and log the real-time sensor data, generating a custom labeled dataset for human activities like walking and sitting.
- Trained a 1D Convolutional Neural Network in PyTorch to classify human activities (walking, sitting), achieving a peak validation accuracy of 87.5% on the test set.
- Visualized training and testing accuracy against learning rate and training time to demonstrate model stability and performance trends, documenting the findings in a comprehensive report.

Sensor Graphic User Interface Arduino

Long Beach, CA

Developer

- Developed a real-time data visualization interface in Python (Tkinter/Matplotlib) for serially streamed MPU9250 accelerometer data from an Arduino microcontroller.
- Engineered a modular project structure including serial communication controllers, GUI logic, and sensor data parsing to enable future reuse across embedded systems projects.
- Validated end-to-end system by integrating sensor readings, asynchronous serial data handling, and python threading for smooth user interaction.

Air Quality Sensor Housing | Engineering Design

Notre Dame, IN

Developer

- Collaborated with a team of four engineering students to develop and iterate on sensor housing with a rack-and-pinion opening mechanism actuated by a servo motor and ESP32-based microcontroller.
- Designed and prototyped with SolidWorks in two design phases, optimizing for form, fit, function, and durability through iterative testing and feedback

LEADERSHIP AND ACTIVITIES

IrishSat's Gravitational Orbital Attitude Thermal Lab

Notre Dame, IN

Electronics Lead

Aug. 2024 – Present

- Appointed as incoming Electronics Lead to oversee embedded systems development and PCB design initiatives for the Helmholtz cage project, coordinating hardware-software integration efforts across the cubesat projects.
- Designed a custom Arduino R3 hat PCB using KiCad to integrate and optimize Arduino, magnetometers, and H-bridge circuits, improving system efficiency and reducing assembly complexity.
- Collaborated under expert mentorship to refine IrishSat's Helmholtz Cage design, integrating PCB design (KiCad), control algorithms (MATLAB), and embedded Linux systems (Raspberry Pi) into a unified testing platform.

TECHNICAL AND LANGUAGE SKILLS

Programming: C, C++, Python, Git, MATLAB

Tools: Microsoft Office Suite, KiCad, Vim, SolidWorks, Fusion 360

Skills: Machine Learning, Technical Writing, Electronics Soldering

Language: English (Native), Spanish (Bilingual)

Hobbies: Japanese(N4-N5), Chess (1000+ Rapid)